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DESS: Probabilistic Sampling in Learned Embedding Spaces

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Abstract Modern AI systems built on learned embedding spaces face a recurring question: once a useful embedding has been learned, how should a model represent uncertainty and diversity inside that space without sacrificing computational tractability? We present DESS (Dimensional-PDFs for Embedding Space Sampling), a parameter-efficient framework that addresses this question by predicting, for each input, the parameters of a per-dimension probability density over the embedding space, and sampling from those densities at inference time. DESS uses simple element-wise losses with no sampling in the training loop, attaches to any embedding-space model with two output heads, and keeps the output set decoupled from the trained predictor.

We validate DESS in three settings of increasing realism. On controlled synthetic Gaussian data we show that DESS recovers ground-truth mean and variance parameters and exposes a controllable spread hyperparameter. On sentence-transformer embeddings of 20 Newsgroups, the predicted per-dimension spread correlates strongly with empirical within-class variance (Spearman $\rho = 0.94$), but the spread norm by itself is not a useful out-of-distribution signal (AUROC 0.49). We show that the behaviour of the spread hyperparameter in high-dimensional embedding spaces is governed by concentration of measure, and turn this observation into a geometric design rule that predicts a safe operating range for the hyperparameter directly from training-set geometry, with no sweep required; the rule transfers to a second dataset within 20% of the empirical threshold. We further show that the learned spread behaves more like a Laplace scale parameter than a Gaussian standard deviation, helping explain why bounded inference-time sampling outperforms Gaussian and Laplace sampling on the same trained model. Finally, we adapt DESS to candidate-list reranking on the SQID/ESCI Task 1 benchmark under the original judged-candidate protocol. The best DESS variant retains the frozen query embedding as the mean and learns only the per-dimension spread around it, matching a strong SBERT-text baseline though not surpassing cross-encoder or text-image fusion baselines. Together these results position DESS as a geometrically principled, calibration-aware uncertainty layer for embedding-space models.

Keywords Probabilistic Sampling · Embedding Spaces · Uncertainty Estimation · Concentration of Measure · Calibration

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1 Introduction

Probabilistic sampling in representation spaces is central to many modern AI systems. Large language models sample tokens from learned distributions to produce varied generations [1], dense retrieval systems benefit from diversity in candidate selection rather than purely deterministic nearest-neighbour lookups [2, 3], and embedding-based recommenders trade exploration against exploitation by varying the sampling temperature of their output distributions [4]. Across these settings, a recurring engineering question arises: once a useful embedding space has been learned, how should a model represent uncertainty and diversity inside that space without sacrificing computational tractability?

This paper presents DESS (Dimensional-PDFs for Embedding Space Sampling), a parameter-efficient framework for probabilistic sampling in learned embedding spaces.[1] Given an input vector, DESS predicts the parameters of a per-dimension probability density over an embedding space and samples from those densities at inference time. The architecture splits the final layer of a backbone network into two heads: a mean head that produces a per-dimension expected value, and a spread head that produces a per-dimension scale. Training uses simple element-wise losses with no sampling in the loop, which keeps the method lightweight and stable. At inference time, the predicted densities are sampled to produce a target vector, and the nearest embedding in the output set is returned. Because the output set $\mathcal{E}$ is consulted only at inference, it can be modified without retraining the predictor, provided new entries remain in the same representation space.

DESS sits in a particular position in the design space of probabilistic neural-network outputs. Mixture density networks [6, 7] model multimodal targets through Gaussian mixtures but are notoriously hard to train in high dimensions. Bayesian neural networks [8, 9] model uncertainty at the level of weights and require sampling or variational approximation at every forward pass. Generative latent-variable models such as VAEs [10] and GANs [11] are designed for sample synthesis rather than selection over an existing embedding set. DESS does not attempt multimodality, weight-level uncertainty, or sample synthesis; it instead aims to provide a lightweight per-dimension uncertainty structure around a point estimate in an existing embedding space. Section 2 develops this comparison in more detail.

We validate DESS in three settings of increasing realism, organized as three experiments. Experiment 1 (Section 4) uses controlled five-class Gaussian data to verify that DESS recovers ground-truth mean and variance parameters and to characterize the role of its spread-controlling hyperparameter. Experiment 2 (Section 5) evaluates DESS on sentence-transformer [12, 13] embeddings of 20 Newsgroups [14]. We chose this dataset because it provides a moderately noisy, semantically structured embedding space with multiple non-trivial classes — enough complexity to exercise the spread parameter without requiring task-specific fine-tuning of the encoder. Experiment 3 (Section 6) adapts DESS to candidate-list reranking on the SQID/ESCI Task 1 benchmark [15, 16], a realistic downstream task with established baselines and a fixed evaluation protocol.

The picture that emerges is mixed in a useful way. DESS reveals meaningful structure in real embedding spaces: predicted spread tracks empirical class-wise variance with Spearman $\rho = 0.94$, and the spread hyperparameter exposes a controllable relevance–diversity tradeoff under sampling. The behaviour of the spread hyperparameter in high-dimensional embedding spaces is governed by concentration of measure [17], and we turn this observation into a geometric design rule that predicts a safe operating range for the hyperparameter directly from training-set geometry, validated on a second dataset [18]. We also show that the learned spread behaves more like a Laplace scale parameter than a Gaussian standard deviation, with consequences for the choice of inference-time sampling distribution. At the same time, the spread norm alone is not an effective out-of-distribution detector [19], and on SQID reranking the strongest DESS variant matches but does not exceed a strong frozen text-embedding baseline.

The aim of the paper is not to claim that DESS dominates every downstream task. It is to present a complete probabilistic-sampling framework for embedding spaces, characterize what its uncertainty structure does and does not capture, and identify the conditions under which it is most useful. In summary, the paper makes the following contributions:

1. We present DESS, a parameter-efficient framework for probabilistic sampling in learned embedding spaces, with a complete description of its formulation, training, and inference (Section 3).
2. We validate DESS on controlled synthetic data, show that it recovers ground-truth distribution parameters, and characterize the role of the spread hyperparameter (Section 4).
3. We evaluate DESS on real sentence-embedding spaces and show that its predicted spread tracks empirical within-class variance, while not serving as an effective out-of-distribution signal (Section 5).
4. We analyze the role of the spread hyperparameter in high-dimensional embedding spaces, derive and validate a geometric design rule for its safe operating range across two datasets, and show that the learned spread is calibration-shaped as a Laplace scale parameter rather than a Gaussian standard deviation (Section 5).
5. We adapt DESS to benchmark-faithful candidate-list reranking on SQID/ESCI and compare several reranker variants under corrected NDCG (Section 6).
6. We discuss what transfers from controlled synthetic conditions to real downstream tasks, and where the

[1] A preliminary version of this work appeared at SGAI 2025 [5]. The present paper extends that work with an analysis of DESS in real learned embedding spaces, a geometric design rule for the spread hyperparameter, a calibration analysis, and an evaluation on the SQID/ESCI reranking benchmark.

method remains competitive or limited (Sections 7 and 8).

[2]

2 Related Work

DESS belongs to a broader family of methods that introduce probabilistic structure into neural-network outputs. We review the most relevant families below and clarify where DESS sits in this landscape.

Mixture density networks (MDNs). Introduced by [6], MDNs predict the parameters of a Gaussian mixture distribution conditioned on the input: mixture weights, component means, and covariances. This allows MDNs to model multimodal output distributions and has made them effective for tasks such as trajectory forecasting and probabilistic face embeddings. However, MDNs are well known to suffer from numerical instability, component collapse, and difficult hyperparameter tuning, particularly in high dimensions [7]. They must also be trained end-to-end with the mixture loss, which prevents straightforward reuse of large pretrained encoders. DESS sidesteps both issues by replacing the mixture formulation with per-dimension unimodal densities and by using simple squared-error and L1 losses rather than a likelihood-based mixture objective. The price of this simplification is that DESS does not represent multimodality.

Bayesian neural networks (BNNs). BNNs treat model parameters as random variables and learn posterior distributions over weights rather than point estimates [8, 9, 20]. This provides a principled treatment of epistemic uncertainty: by sampling weight configurations from the posterior, a BNN can produce a distribution of outputs for a single input. The cost is computational: storing and updating distributions over every weight is expensive, and sampling at inference time multiplies the forward-pass cost by the number of Monte Carlo samples. BNNs scale poorly to large architectures or high-dimensional outputs. DESS expresses uncertainty inside the embedding space (over outputs) rather than over weights, and runs at the cost of a standard feed-forward network with two output heads.

[2] A preliminary version of this work appeared at SGAI 2025 [5]. The present paper extends that work with an analysis of DESS in real learned embedding spaces, a geometric design rule for the spread hyperparameter, a calibration analysis, and an evaluation on the SQID/ESCI reranking benchmark.

Generative latent-variable models. VAEs [10] and GANs [11] are designed primarily to learn complex data distributions for synthesis. A VAE jointly trains a probabilistic encoder and decoder by maximizing the evidence lower bound, and uses the decoder as a generative model by sampling from a prior. A GAN trains a generator against a discriminator in a two-player game, learning an implicit density that supports sampling but not likelihood evaluation. While powerful for unconditional generation, neither family is well suited for probabilistic selection over an existing set of embeddings: VAEs require a specific encoder–decoder architecture and stochastic sampling during training, and GANs do not provide tractable likelihoods. DESS is intentionally designed for the selection setting and avoids both constraints.

Categorical-likelihood models. A dominant paradigm in NLP and sequential recommendation frames the prediction task as selection from a fixed discrete vocabulary using a softmax output layer over a learned dictionary. Autoregressive language models [**?** **?**] and softmax-based recommenders [4] share this mechanism. Despite strong empirical performance, the categorical formulation is rigid: the output dictionary must be fixed at training time, and adding tokens or items requires resizing the output layer and retraining. The number of output tokens directly determines the size and computational cost of the final layer, and tokenization choices have been shown to introduce fairness disparities across languages [**?**]. DESS expresses the prediction over a continuous embedding space rather than a discrete dictionary, which decouples the trained predictor from the output set: the set of available embeddings can change at inference time without retraining.

Calibration of probabilistic predictions. A separate line of work studies whether predicted probabilities reflect empirical frequencies, which is closely related to our analysis of the DESS spread parameter. [21] popularized the term calibration for modern neural networks and introduced reliability diagrams as a diagnostic tool. The aleatoric–epistemic decomposition [20, 22] distinguishes uncertainty due to inherent noise from uncertainty due to limited data or model capacity. We use this distinction in Section 5 to clarify that DESS's spread parameter, while well-aligned with within-class variance, does not function as an epistemic-uncertainty signal. The Laplace-versus-Gaussian calibration analysis we present in Section 5.6 draws on classical maximum-likelihood theory [23, 24] for L1-loss estimators.

Concentration of measure in high dimensions. A recurring theme in our analysis is that low-dimensional in-

tuitions about probability density behave differently in high dimensions [17]. The mass of a high-dimensional Gaussian concentrates on a thin shell at distance approximately $\sqrt{d} \cdot \sigma$ from its mean, which has direct consequences for sampling-based retrieval: even modest per-dimension spread produces sample displacements that exceed inter-class distances in the embedding space. We use this observation in Section 5.4 to explain why a hyperparameter setting that performs well in two dimensions performs poorly in several hundred, and use it as the basis for a geometric design rule.

Positioning summary. DESS occupies a distinct position relative to these families: it provides per-dimension uncertainty over embedding-space outputs (unlike BNNs over weights), without multimodality (unlike MDNs), without sample synthesis (unlike VAEs and GANs), and without a fixed output dictionary (unlike categorical softmax models). The trade-off is deliberate: DESS aims to be a lightweight, retrofittable layer that adds a calibrated uncertainty structure to an existing embedding model, not a replacement for any of the families above. The remainder of the paper validates this positioning empirically.

3 The DESS Method

This section presents DESS in full: the problem setting it addresses, the form of the predicted density, the architecture, the training losses, and inference. Each design choice is motivated, and the formulation is consistent with the computational graph shown in Figure **??**.

3.1 Problem Setting

Let $\mathcal{E} = \{e_1, \ldots, e_n\} \subset \mathbb{R}^d$ be a fixed set of embeddings representing structured outputs — for example, the text embeddings of all documents in a 20 Newsgroups class (Section 5) or the candidate products attached to a query in the SQID reranking benchmark (Section 6). Given an input vector x, the task is to select an embedding $e_i \in \mathcal{E}$ appropriate for the input.

A naive approach treats this as direct classification over $\mathcal{E}$: the model produces a categorical distribution over the n embeddings and selects via arg max. This approach is conceptually simple but has two limitations that motivate DESS. First, the output dimension grows with $|\mathcal{E}|$, which makes the final layer expensive when the embedding set is large. Second, the trained predictor is bound to a specific output dictionary: changing $\mathcal{E}$ requires resizing the output layer and retraining.

DESS instead formulates the task as probabilistic sampling in $\mathbb{R}^d$. The model defines a conditional density $p(y \mid x)$ on the embedding space, draws a sample $\hat{y} \sim p(y \mid x)$, and returns the embedding $e_j \in \mathcal{E}$ closest to $\hat{y}$ in Euclidean distance:

$$\hat{e} = \arg\min_{e \in \mathcal{E}} \|\hat{y} - e\|_2. \tag{1}$$

This reframing has two consequences that matter throughout the paper. First, the output set is decoupled from the trained predictor: $\mathcal{E}$ can be modified at inference time without retraining, provided new entries remain in the same representation space. Second, the density $p(y \mid x)$ is itself an object of interest; its spread parameter carries information about the range of plausible outputs for a given input, which we exploit in Sections 5.4–5.7.

3.2 Per-Dimension Independent Densities

DESS defines $p(y \mid x)$ as a product of per-dimension univariate densities:

$$p(y \mid x) = \prod_{i=1}^{d} p_i(y_i \mid x). \tag{2}$$

In the present formulation each p_i is a univariate Gaussian $\mathcal{N}(\hat{\mu}_i, \hat{\sigma}_i^2)$, parameterized by two d-dimensional outputs of the network: a mean vector $\hat{\boldsymbol{\mu}}$ and a spread vector $\hat{\boldsymbol{\sigma}}$, both indicated in Figure **??**.

The per-dimension factorization is a deliberate simplification. A full covariance model could express dependencies between dimensions but would require predicting $O(d^2)$ parameters and would complicate training. Per-dimension independence trades that expressiveness for stable training with explicit element-wise targets and no sampling in the loop. As we show empirically (Section 5.3), the resulting spread vector still tracks empirical class-conditional variance closely, which suggests that diagonal independence is a useful approximation in the embedding spaces we study.

3.3 Architecture

The DESS architecture is a feedforward network whose final layer is split into two heads of dimension d each. The mean head produces $\hat{\boldsymbol{\mu}}$ and the spread head produces $\hat{\boldsymbol{\sigma}}$ (Figure **??**). The spread head passes through a softplus activation with a small positive floor to guarantee strictly positive outputs — a requirement for $\hat{\boldsymbol{\sigma}}$ to serve as a scale parameter regardless of the inference-time distribution interpretation (see Section 5.7).

Any backbone can feed the two heads. The contribution of DESS is not the backbone but the loss structure on the two heads.

3.4 Training Losses

DESS is trained with element-wise losses on $\hat{\boldsymbol{\mu}}$ and $\hat{\boldsymbol{\sigma}}$ separately. No sampling is performed during training. We describe two cases.

Single-target case. When each input x has a single target embedding $y \in \mathbb{R}^d$, the mean head is trained with a squared-error loss

$$\mathcal{L}_\mu = (\hat{\boldsymbol{\mu}} - y)^2, \tag{3}$$

applied element-wise. This is standard regression of $\hat{\boldsymbol{\mu}}$ onto y.

The spread head is trained toward a target derived from the *current* mean-prediction error, rather than from any explicit ground-truth spread. The motivation is that during training we typically do not have access to a ground-truth distribution from which y was sampled; we only observe one y per input. The mean residual provides a proxy for spread that is accessible at every training step. Specifically, let $\delta = |\hat{\boldsymbol{\mu}} - y|$ denote the element-wise absolute error of the mean head. The spread target is $\beta \cdot \delta$ where $\beta > 0$ is a scalar hyperparameter, and the spread loss is

$$\mathcal{L}_\sigma = \big|\hat{\boldsymbol{\sigma}} - \beta \cdot \delta\big|. \tag{4}$$

The hyperparameter β controls how aggressively $\hat{\boldsymbol{\sigma}}$ tracks the residual: $\beta = 1$ would have $\hat{\boldsymbol{\sigma}}$ match the absolute residual exactly, while larger β produces broader predicted distributions. Section 4.2 characterizes the role of β in detail, and Section 5.5 derives a design rule for choosing it in high-dimensional embedding spaces.

A crucial design choice is that δ is treated as a constant target when computing $\mathcal{L}_\sigma$: gradients do not flow from $\mathcal{L}_\sigma$ back through the mean head (the stop-gradient block in Figure **??**). Without this stop-gradient, the spread loss could shape $\hat{\boldsymbol{\mu}}$ to make δ easier to predict, distorting the mean head away from its regression target. The stop-gradient enforces a clear division of labor: the mean head learns y via $\mathcal{L}_\mu$, and the spread head reports the mean head's errors honestly via $\mathcal{L}_\sigma$.

Multi-target case. The single-target case assumes one target embedding per input. In many practical settings each input is associated with a *set* of plausible target embeddings rather than a unique one. For example, in our text-embedding experiments (Section 5) the input is a one-hot class label and the targets are the document embeddings within that class. There are many such embeddings, and the spread of $\hat{\boldsymbol{\sigma}}$ should reflect the spread of the targets within the class, not just the residual against a single arbitrary one of them.

We extend the loss to handle this case. Given T target embeddings $y_1, \ldots, y_T$ for a single input x, the mean head is trained against the averaged target $\bar{y} = \frac{1}{T}\sum_t y_t$ using the same squared-error loss as in Equation (3). The spread target combines two signals: the residual $\delta = |\hat{\boldsymbol{\mu}} - \bar{y}|$ (as before) and a per-dimension dispersion vector capturing how the targets themselves vary,

$$\Delta y = \max_t y_t - \min_t y_t. \tag{5}$$

The two signals are combined with a weighting $\alpha \in [0, 1]$ and the scaling factor β:

$$\tau = \beta\big(\alpha \cdot \delta + (1 - \alpha) \cdot \Delta y\big), \tag{6}$$

and the spread loss is $\mathcal{L}_\sigma = |\hat{\boldsymbol{\sigma}} - \tau|$. The α parameter interpolates between two regimes: at $\alpha = 1$ the loss reduces to the single-target case (only the mean residual matters), and at $\alpha = 0$ the loss is driven entirely by the observed dispersion of the target set. In all multi-target experiments below we set $\alpha = 0.5$, which gives equal weight to both signals.

The two losses combine additively into the total training loss $\mathcal{L} = \mathcal{L}_\mu + \mathcal{L}_\sigma$ (Figure **??**).

3.5 Inference

At inference time, the model produces $\hat{\boldsymbol{\mu}}$ and $\hat{\boldsymbol{\sigma}}$ for a given input x, and a prediction is constructed by sampling one value per dimension,

$$\hat{y}_i \sim \mathcal{N}(\hat{\mu}_i, \hat{\sigma}_i^2), \qquad i = 1, \ldots, d, \tag{7}$$

followed by nearest-neighbour selection in $\mathcal{E}$ via Equation (1). A point-estimate variant uses $\hat{\boldsymbol{\mu}}$ directly as $\hat{y}$, skipping the sampling step. We treat the sampling variant as the primary use of DESS, and the point-estimate variant as a diagnostic that isolates the contribution of the mean head from the spread head.

Note on normalization. The formulation makes no assumption about whether embeddings in $\mathcal{E}$ are normalized. In our text-embedding experiments (Section 5), $\mathcal{E}$ consists of L2-normalized vectors on the unit sphere, but the sampled vector $\hat{y}$ produced by Equation (7) is not itself constrained to the sphere. We do not reproject $\hat{y}$ before nearest-neighbour retrieval; the retrieval is performed in the ambient space $\mathbb{R}^d$ under Euclidean distance. This is a deliberate modeling choice rather than an incidental one, since the geometric analysis in Section 5.4 depends on it.

3.6 The Role of the Spread Hyperparameter

The scalar β controls the scale of the learned spread. Following Equation (4), at convergence $\hat{\boldsymbol{\sigma}}$ tracks β times the mean head's residual error. Larger β produces broader sampling distributions and more diverse outputs at inference; smaller β produces tighter distributions closer to the point estimate.

The role of β in the original synthetic experiments (Section 4) is straightforward: a value of $\beta = 1.5$ emerges as optimal in two-dimensional Gaussian targets. Whether this choice transfers to higher-dimensional embedding spaces is a question we return to in Section 5.4, where we show that it does not, and in Section 5.5, where we derive a geometric replacement.

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4 Experiment 1: Synthetic Validation

The first experiment validates DESS on controlled synthetic Gaussian data. The goal is to test whether the method recovers known distributional parameters, to characterize the role of the spread hyperparameter β in a setting where ground truth is available, and to surface methodological considerations that affect spread estimation in practice. Using synthetic data here is deliberate: it provides a clean testbed where the target distributions are known and where any deviation between predicted and ground-truth parameters can be attributed to the method rather than to encoder artifacts. The findings of this experiment also motivate the choices we make in Experiments 2 and 3, including the choice of $\beta = 1.5$ as a starting point and the use of L1 loss on the spread head.

4.1 Setup

Data. We construct a five-class synthetic dataset where each class has 100 training samples drawn from a class-specific 2D Gaussian. The means and standard deviations are summarized in Table 1. Four classes have fixed parameters; the fifth class (centered at $[0.5, 0.5]$) has its standard deviation varied across $\sigma \in \{0.05, 0.1, 0.2, 0.3\}$ to test DESS's adaptation to different spread regimes. All samples are clipped to the unit square $[0, 1]^2$ to mimic dense embedding components, which matters for the analysis in Section 4.3. Figure 1 shows the resulting datasets.

Model. We instantiate DESS with a three-layer MLP whose input is a 5-dimensional one-hot class label and whose output is a 4-dimensional vector concatenating $\hat{\boldsymbol{\mu}} \in \mathbb{R}^2$ and $\hat{\boldsymbol{\sigma}} \in \mathbb{R}^2$. The hidden layer dimensions are $5 \to 32 \to 32 \to 4$. We train with the single-target loss of Section 3.4 (one target y sampled per input at each step), using Adam [25] with learning rate 10^{-3} and batch size 32, for 100 epochs.

Evaluation. We compare the model's predicted $\hat{\boldsymbol{\mu}}$ and $\hat{\boldsymbol{\sigma}}$ against the known ground-truth parameters used to generate each class. We focus particular attention on the spread error for the variable-spread fifth class, $\left|\hat{\sigma} - \sigma_{\text{true}}\right|$, as a function of the spread hyperparameter β. For each combination of target σ and $\beta \in \{0.5, 1.0, 1.25, 1.5, 1.75, 2.0, 2$ we train with a fixed random seed for reproducibility.

Table 1 Gaussian parameters for the synthetic five-class dataset. The fifth class varies its standard deviation across four values to test DESS's ability to learn different spread regimes. Each covariance matrix is diagonal.

Class label	Mean μ	Std. dev. (σ_x, σ_y)
$[1, 0, 0, 0, 0]$	$[0.7, 0.2]$	$[0.08, 0.12]$
$[0, 1, 0, 0, 0]$	$[0.2, 0.3]$	$[0.10, 0.15]$
$[0, 0, 1, 0, 0]$	$[0.7, 0.6]$	$[0.12, 0.18]$
$[0, 0, 0, 1, 0]$	$[0.2, 0.6]$	$[0.15, 0.20]$
$[0, 0, 0, 0, 1]$	$[0.5, 0.5]$	$[\sigma, \sigma]$

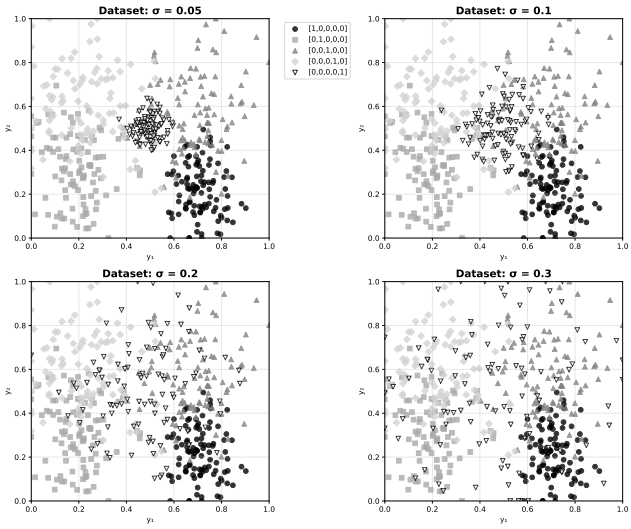

Fig. 1 Synthetic five-class datasets at four values of the variable spread σ. Each subplot shows the distribution of 100 training samples per class. The first four classes maintain fixed parameters across all four datasets; the fifth class (center) varies its standard deviation across $\sigma \in \{0.05, 0.1, 0.2, 0.3\}$. All samples are clipped to the unit square $[0, 1]^2$.

4.2 Hyperparameter Analysis

The systematic β sweep yields a clear pattern: $\beta = 1.5$ consistently minimizes spread error across all four target σ values (Figure 2). Both smaller values (which

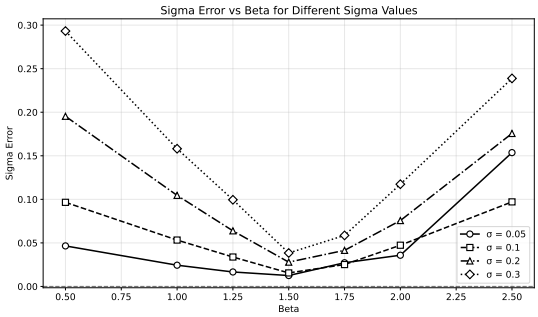

Fig. 2 Sigma error as a function of β for different target σ values on the synthetic five-class dataset. Lower values indicate better spread estimation. The optimal $\beta = 1.5$ consistently minimizes sigma error across all target distribution spreads.

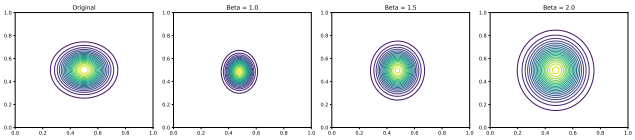

Fig. 3 Learned distributions for varying β values compared to the ground-truth distribution on the synthetic five-class dataset. The contour plots show how β affects spread estimation, with $\beta = 1.5$ providing the best approximation to the target distribution.

under-estimate the target spread) and larger values (which over-estimate it) produce higher error. The minimum at $\beta = 1.5$ is robust across the four target distributions tested, suggesting it is a useful default for low-dimensional Gaussian targets.

The visual effect of β on the learned distributions is shown in Figure 3. At $\beta = 1.0$ the learned distribution is too tight relative to the ground truth; at $\beta = 2.0$ it is too broad; at $\beta = 1.5$ it closely matches. Beyond simply controlling fit quality, this gives the practitioner a clear lever for the relevance–diversity tradeoff under sampling: smaller β concentrates samples near the predicted mean (favoring relevance), while larger β spreads them more widely (favoring diversity). We exploit this lever in Section 5.4.

We note one caveat that becomes important later. The result $\beta = 1.5$ is specific to this two-dimensional setting. In high-dimensional embedding spaces, concentration of measure changes the relationship between β and the resulting sample displacement, and a different β regime applies (Section 5.4).

4.3 Training Dynamics and Sampling Bias

DESS converges quickly on this task. Both the mean and spread parameters reach their target values within approximately 20 epochs, with the spread parameters showing additional gradual refinement through the remainder of training. After convergence, the parameters remain stable with minimal oscillation, indicating that the dual-loss training is well-behaved despite the asymmetric stop-gradient structure (Section 3.4).

A methodological caveat that emerged during this experiment is worth recording. Our synthetic data is generated by sampling from Gaussian distributions and clipping to $[0, 1]^2$, which mimics the bounded component values of L2-normalized embeddings. As the target σ increases, more samples fall outside the unit square and get clipped, which systematically reduces the empirical standard deviation of the training data below the intended sampling σ. The model's spread predictions therefore appear to undershoot the target σ at higher spread levels, but they actually match the empirical spread of the training data they observed. We mention this here because clipping bias is a common source of confusion in synthetic spread-estimation experiments and because it reinforces the importance of computing empirical statistics from the actual training data rather than from the generative parameters when interpreting DESS's output.

4.4 Summary

The synthetic experiment establishes three things that we rely on in the remainder of the paper. First, DESS recovers ground-truth distributional parameters when the data is well-controlled, confirming that the formulation in Section 3 is sound. Second, the spread hyperparameter β has a clear and predictable effect on the learned distributions: $\beta = 1.5$ is optimal in two dimensions, and β provides an intuitive control over the relevance–diversity tradeoff under sampling. Third, training is stable and converges quickly, even with the dual-loss structure and the stop-gradient on the spread target. The remainder of the paper asks whether and how these properties carry over to real learned embedding spaces.

5 Experiment 2: DESS in Text Embedding Spaces

Experiment 1 established that DESS recovers known distributional parameters in a controlled two-dimensional Gaussian setting. The natural next question is whether the method's predicted spread carries meaningful information when the embedding space is itself learned, and whether the hyperparameter choices that work well in low dimensions transfer to the high-dimensional embed-

ding spaces typical of modern representation learning. This experiment addresses both questions.

5.1 Setup

We study DESS on the 20 Newsgroups corpus [14], encoded with sentence-transformer embeddings. We chose this dataset for three reasons. First, it provides a moderately noisy, semantically structured embedding space with twenty non-trivial classes spanning a range of topics — enough complexity to exercise the spread parameter without requiring task-specific fine-tuning of the encoder. Second, the canonical splits and class structure are well established in the literature, which makes classification accuracy a meaningful diagnostic even though state-of-the-art accuracy is not our objective. Third, the dataset is small enough that we can train and evaluate many DESS configurations without prohibitive compute, which matters for the systematic β-sweep analyses in Sections 5.4 and 5.5. The goal of this experiment is not to set a new benchmark on 20 Newsgroups but to use the dataset as a well-understood probe for what DESS's spread parameter learns.

Data. We use the full 20 Newsgroups corpus with headers, footers, and quoted text removed to prevent metadata leakage across documents. Of the twenty topic classes, we hold out five (`sci.crypt`, `rec.sport.hockey`, `talk.politics.mideast`, `misc.forsale`, `alt.atheism`) as out-of-distribution for the OOD-detection analysis in Section 5.3; these particular classes were chosen to span technical, recreational, political, commercial, and ideological topics so that out-of-distribution evaluation is not dominated by topical similarity to a single in-distribution class. The remaining fifteen classes are used for training and evaluation, split 60/20/20 into train/validation/test, stratified by class.

Embeddings. Each document is encoded with `all-MiniLM-L6-v2` [12, 13], a sentence-transformer producing 384-dimensional vectors that we L2-normalize to unit norm. We verified that embedding norms are exactly 1.0 and that the mean per-dimension standard deviation across the training set is 0.0497, placing all subsequent analyses on the unit sphere.

Model. Following the multi-target spirit of DESS (Section 3.4), we train a class-conditional model that maps a one-hot class label to a distribution over the corresponding text-embedding space. The architecture is a three-layer MLP with hidden dimensions $(256, 256)$, ReLU activations, and dropout 0.1 [26], with a final split into mean and spread heads of size 384 each. The spread head passes through a softplus activation with a small floor to guarantee strict positivity regardless of the inference-time distribution interpretation (Section 5.7).

Table 2 Classification results on 20 Newsgroups over five seeds, mean ± std. The regression baseline is DESS with the spread head removed — an ablation that isolates the contribution of the spread head to classification accuracy.

Method	Accuracy	Macro-F1
DESS	0.6999 ± 0.0013	0.6937 ± 0.0014
Regression baseline	0.7025 ± 0.0015	0.6959 ± 0.0014
Nearest-centroid floor	0.7016	0.6957
Softmax baseline	0.7289 ± 0.0047	0.7200 ± 0.0064

Training. We use the multi-target loss (Equation 6) with $\alpha = 0.5$ and (unless otherwise specified) $\beta = 1.5$ — the value identified as optimal in the synthetic experiments of Section 4. For each training step we sample a class c uniformly at random, take $T = 16$ of its training embeddings as multi-target, and compute the squared-error loss on $\hat{\boldsymbol{\mu}}$ and the L1 loss on $\hat{\boldsymbol{\sigma}}$ against the appropriately scaled target. We train with Adam [25] at learning rate 10^{-3}, batch size 64, for 100 epochs.

Baselines. We compare DESS against three baselines:

1. a *regression baseline* that predicts only the mean (i.e., DESS without the spread head) and retrieves by nearest mean,
2. a *softmax classifier* trained with cross-entropy directly on SBERT features, and
3. a *nearest-centroid floor* using class centroids in the embedding space.

The comparison is intentionally revealing rather than a competition. If DESS underperforms a strong discriminative classifier on classification accuracy, that does not invalidate the method — DESS is solving a richer problem (modeling per-class distributions in embedding space) than direct label prediction. The relevant question is whether the probabilistic structure DESS learns is useful and interpretable, which the subsequent subsections address.

5.2 Classification and Representation Quality

Table 2 reports classification results over five seeds. DESS is competitive with simple embedding-based baselines, but does not outperform the softmax upper bound.

These results support a careful interpretation. DESS is not most valuable when judged purely as a direct

accuracy-optimizing classifier; on this task a conventional softmax model remains stronger. However, DESS remains competitive with embedding-based point-estimate methods while providing something those methods do not: an explicit uncertainty structure in the embedding space. The value of DESS is best understood not through a single accuracy number but as a mechanism for modeling structured variation, which the following subsections examine in detail.

5.3 Is the Predicted Spread Meaningful?

To test whether the predicted spread parameter captures meaningful information, we compare $\hat{\boldsymbol{\sigma}}$ to the empirical per-dimension variance of the training embeddings within each class. The Spearman correlation between predicted spread and empirical standard deviation, across all (class, dimension) pairs, is $\rho = 0.9395$ (Figure 4). This is one of the strongest findings in the paper: DESS does not produce arbitrary spread magnitudes but predicted uncertainty that aligns closely with the actual geometry of class-conditioned variation in the embedding space. Figure 4 also reveals that the relationship is not the identity: the point cloud lies along a line with a larger slope than $y = x$, reflecting that the learned $\hat{\boldsymbol{\sigma}}$ is systematically scaled up relative to the empirical standard deviation. Section 5.6 interprets this scaling via the L1 loss and the β multiplier.

The spread parameter is less useful for detecting novelty, however. When we use the norm of the predicted spread as an out-of-distribution score [19] on the five held-out classes, the resulting AUROC is 0.4907 indistinguishable from chance. An independent check using $\|\hat{\boldsymbol{\sigma}}\|$ as a selective-classification signal confirms this: OOD rejection rates do not systematically differ from in-distribution rejection rates at matched confidence thresholds.

This asymmetry clarifies what the DESS spread parameter should and should not be interpreted as. In this real-data setting it is a good descriptor of *aleatoric* variation within a learned embedding cluster, but not a reliable detector of *epistemic* novelty [20, 22]. The distinction matters: if DESS is to support out-of-distribution detection or abstention, it likely needs an explicit uncertainty objective or an auxiliary calibration mechanism rather than relying on spread magnitude alone.

5.4 Sampled Classification and Concentration of Measure

We now examine how the spread hyperparameter β affects downstream behavior under sampling. As β increases from 0.5 to 2.5, sampled classification accuracy decreases monotonically from 0.5480 to 0.0704, while mean intra-list diversity (mean pairwise L2 distance among fifty samples drawn per input) increases from 1.2097 to 6.0678 (Figure 5). An L2 diversity of 6.07 substantially exceeds the diameter of the unit sphere (2.0), indicating that most high-β samples lie outside the embedding manifold entirely — a point we explain below. Relative to the point-estimate accuracy of 0.70 in Table 2, even the smallest tested $\beta = 0.5$ loses roughly fifteen accuracy points under sampling. This result is initially surprising: $\beta = 1.5$ was identified as optimal in Experiment 1 (Section 4), yet here it reduces sampled accuracy to 0.16.

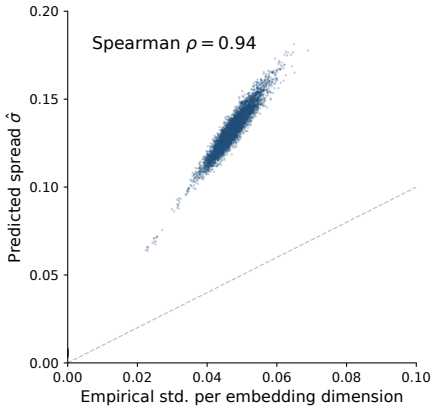

Fig. 4 Predicted spread $\hat{\boldsymbol{\sigma}}$ versus empirical per-dimension standard deviation across all (class, dimension) pairs on 20 Newsgroups. Each point is one of the 384 embedding dimensions for one of the 15 in-distribution classes. The dashed line is the identity $y = x$. The tight alignment of the point cloud (Spearman $\rho = 0.94$) indicates that DESS does not produce arbitrary spread magnitudes; predicted uncertainty tracks the geometry of class-conditioned variation in the embedding space. The cloud lies well above the identity line, reflecting a systematic overestimation of spread analyzed in Section 5.6.

The mechanism behind this collapse is concentration of measure [17]. For a sample $\hat{y} \sim \mathcal{N}(\hat{\boldsymbol{\mu}}, \hat{\boldsymbol{\sigma}}^2 I)$ in d-dimensional space, the Euclidean displacement $\|\hat{y} - \hat{\boldsymbol{\mu}}\|_2$ concentrates sharply around $\sqrt{d} \cdot \bar{\sigma}$, where $\bar{\sigma}$ is the mean per-dimension spread. In our setting $d = 384$ and typical per-dimension $\bar{\sigma}$ grows linearly with β, so expected sample displacement grows as $\beta \cdot \sqrt{d} \cdot \bar{\sigma}_{\text{emp}}$. Empirically, mean sample displacement matches $\|\hat{\boldsymbol{\sigma}}\|$ to three decimal places across all tested β values (e.g., 0.857 vs. 0.857 at $\beta = 0.5$; 2.589 vs. 2.591 at $\beta = 1.5$), confirming that the concentration prediction holds. Meanwhile the minimum pairwise distance between class centroids in our embedding space is only 0.18. Once $\|\hat{\boldsymbol{\sigma}}\|$ exceeds this minimum, samples routinely fall closer to the centroid of a different class than to their own, and nearest-class retrieval breaks. The right panel of Figure 5 makes

this concrete: at $\beta \geq 1.0$, mean intra-list L2 diversity exceeds the unit-sphere diameter, confirming that samples at the synthetically-optimal β fall outside the embedding manifold.

This is not a failure of DESS but a consequence of translating low-dimensional intuition about β into a high-dimensional embedding space. Experiment 1 used $d = 2$, where $\sqrt{d} \cdot \bar{\sigma}$ remains small. In realistic embedding dimensionalities, the same β produces vastly larger L2 displacements. The correct β for a given dataset must be chosen with this scaling in mind, which motivates the design rule we develop next.

5.5 Predicting the Safe Spread Regime from Embedding Geometry

The previous subsection identified concentration of measure as the mechanism behind the β-driven collapse and suggested that the safe operating range of β should be governed by the geometry of the target embedding space. Here we turn this observation into a predictive rule and validate it on a second dataset.

Deriving the rule. For nearest-class retrieval to preserve the correct class assignment under sampling, a sample $\hat{y}$ drawn around $\hat{\boldsymbol{\mu}}_c$ must remain closer to $\hat{\boldsymbol{\mu}}_c$ than to any other class centroid. Combining this with the concentration result $\|\hat{y} - \hat{\boldsymbol{\mu}}_c\| \approx \|\hat{\boldsymbol{\sigma}}_c\|$, and approximating $\|\hat{\boldsymbol{\sigma}}_c\| \approx \beta \cdot \sqrt{d} \cdot \bar{\sigma}_{\text{emp}}$, we obtain the design rule

$$\beta_{\max} \approx \frac{d_{\min}}{\sqrt{d} \cdot \bar{\sigma}_{\text{emp}}}, \tag{8}$$

where $d_{\min}$ is the minimum pairwise distance between class centroids and $\bar{\sigma}_{\text{emp}}$ is the mean per-dimension empirical standard deviation averaged across classes.

The approximation $\|\hat{\boldsymbol{\sigma}}_c\| \approx \beta \cdot \sqrt{d} \cdot \bar{\sigma}_{\text{emp}}$ rests on two assumptions. First, under squared-error training of the mean head with sufficient capacity, the per-dimension residual $|\hat{\boldsymbol{\mu}} - y|$ at convergence tracks the per-dimension empirical standard deviation of the class — since squared-error regression recovers the conditional mean and the residual variance equals the within-class variance [24]. Second, the single-target spread loss $\mathcal{L}_\sigma = |\hat{\boldsymbol{\sigma}} - \beta \cdot \delta|$ then drives $\hat{\boldsymbol{\sigma}}$ toward $\beta \cdot |\hat{\boldsymbol{\mu}} - y|$, so per-dimension $\bar{\sigma}$ tracks $\beta \cdot \bar{\sigma}_{\text{emp}}$ and the concentration result gives the stated approximation.

The rule is derived under the single-target case, but our experiments use multi-target training with $\alpha = 0.5$. The multi-target loss reduces to single-target behavior in the $\alpha \to 1$ limit, and the dispersion term Δy adds positive scale on top of the single-target prediction; we therefore expect the rule to be *conservative* (predicting safe regimes that are, if anything, slightly tighter than the true safe regime) when applied to multi-target training, which is the behavior we observe empirically below. The right-hand side of Equation 8 depends only on training-set geometry and can be computed *before* any DESS model is trained.

Table 3 AG News transfer validation of the design rule. Predicted $\beta_{\max} = 0.31$; the empirical cliff occurs between $\beta = 0.25$ (safe) and $\beta = 0.50$ (unsafe).

β	$\|\hat{\boldsymbol{\sigma}}\|$	Accuracy	Safe?
0.05	0.090	0.875	✓
0.10	0.183	0.872	✓
0.25	0.456	0.854	✓
0.50	0.901	0.772	×
1.00	1.771	0.423	×
1.50	2.642	0.342	×
2.00	3.544	0.367	×
3.00	5.304	0.359	×

Transfer validation. We test whether the rule transfers to a dataset different from the one that motivated it. AG News [18] is a four-class news topic classification benchmark; we encode its documents with the same SBERT model as before and compute its embedding-space geometry. The rule predicts $\beta_{\max} \approx 0.31$ from training data alone. We then train DESS at $\beta \in \{0.05, 0.1, 0.25, 0.5, 1.0$ on AG News and measure sampled accuracy at each β. Figure 6 and Table 3 show the result. Sampled accuracy stays within 0.05 of the point-estimate accuracy (0.872) for $\beta \leq 0.25$ and drops sharply thereafter. The empirical $\beta_{\max}$ (the largest tested β preserving accuracy within tolerance) is 0.25, within 20% of the predicted value. Counting disagreements between rule prediction and empirical safety across all tested β, concordance is 87.5%; the one disagreement (the rule calls $\beta = 0.25$ unsafe but accuracy holds) is conservative, i.e., a false negative on safety rather than a false positive.

Practical implication. For practitioners deploying DESS on a new embedding space, Equation 8 provides a concrete starting point for β selection that requires no training sweep. The rule is conservative in the sense that it predicts safe β ranges; larger β may still be appropriate for applications (such as diverse retrieval) that tolerate some accuracy loss in exchange for sample variability.

This result also retroactively explains why $\beta = 1.5$ from the synthetic experiment of Section 4 performs poorly at $d = 384$: the rule predicts $\beta_{\max} \approx 0.09$ on 20 Newsgroups, so $\beta = 1.5$ operates roughly 16× above the safe regime in this setting.

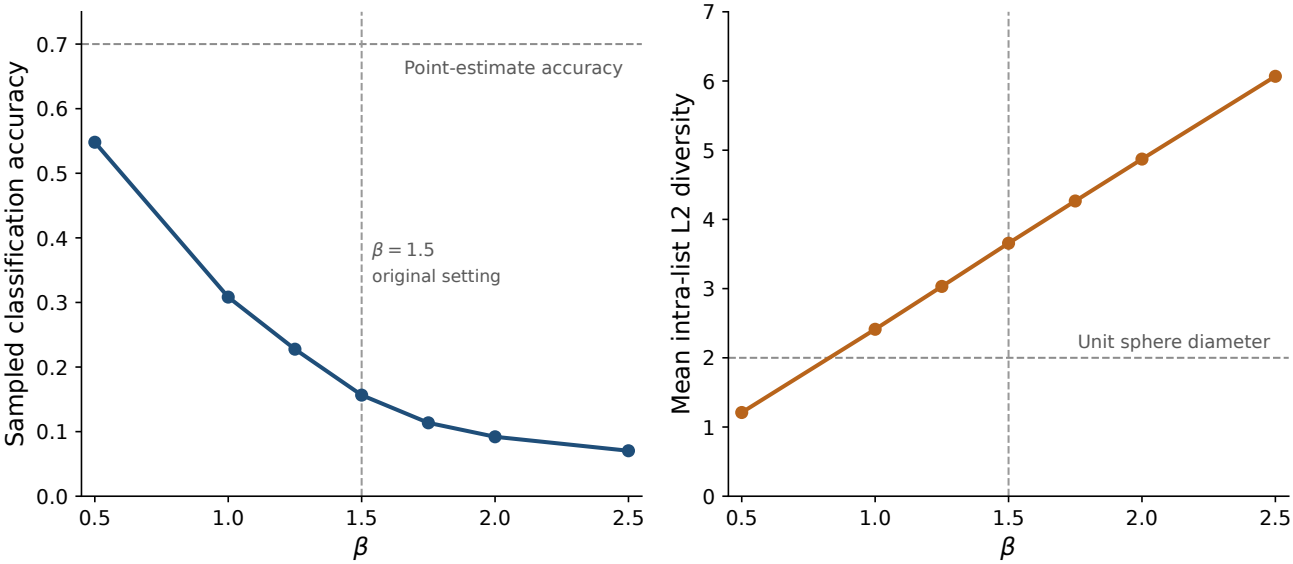

Fig. 5 Effect of the DESS spread hyperparameter β on downstream behavior under sampling, on 20 Newsgroups. *Left:* sampled classification accuracy decreases monotonically with β, falling well below the point-estimate accuracy of 0.70 (horizontal dashed line) even at the smallest tested value. The vertical dashed line marks $\beta = 1.5$, identified as optimal in the synthetic experiments of Section 4. *Right:* mean pairwise L2 diversity among fifty samples drawn per input increases monotonically with β. The horizontal dashed line at 2.0 marks the diameter of the unit sphere on which the embeddings lie; diversities above this line indicate that samples routinely fall outside the embedding manifold, consistent with the concentration-of-measure analysis in the text.

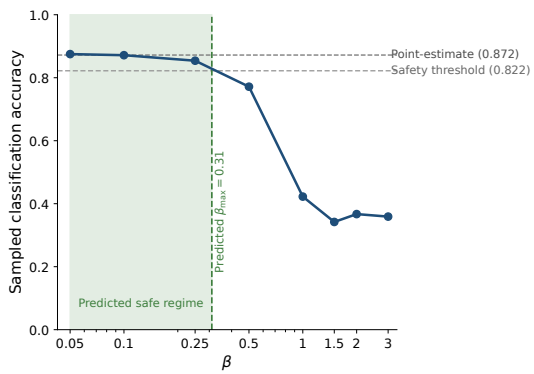

Fig. 6 Design-rule validation on AG News. Sampled classification accuracy is measured across a β sweep (markers, blue curve). Horizontal dashed lines mark the point-estimate accuracy (0.872) and the safety threshold (point estimate minus 0.05 tolerance, 0.822). The vertical dashed line at $\beta = 0.31$ is the $\beta_{\max}$ predicted from training-set geometry alone via Equation 8, before any DESS model is trained. The shaded region indicates the predicted safe regime. Empirical accuracy is preserved within tolerance for $\beta \leq 0.25$ and collapses for $\beta \geq 0.50$, consistent with the rule's prediction to within 20%.

5.6 Calibration of the Spread Parameter

The previous subsections establish that the spread parameter is rank-correlated with empirical variance and that its overall magnitude is geometrically constrained. An additional question is whether $\hat{\boldsymbol{\sigma}}$ is *probabilistically calibrated* [21]: if $\hat{\boldsymbol{\sigma}}$ reports a particular value for a particular (class, dimension), does the corresponding empirical distribution of test-point deviations actually concentrate at that scale?

We answer this by sweeping across nominal coverage levels $\alpha \in \{0.1, 0.2, \ldots, 0.9, 0.95, 0.99\}$. For each α we compute the fraction of (class, dimension, test point) triples for which the test-point deviation from $\hat{\boldsymbol{\mu}}$ falls within the nominal α-interval under two distributional interpretations of $\hat{\boldsymbol{\sigma}}$: Gaussian (where the interval is $\hat{\mu} \pm z_{\alpha/2} \cdot \hat{\sigma}$) and Laplace (where it is $\hat{\mu} \pm \hat{\sigma} \cdot \ln(1/(1-\alpha))$). Perfect calibration corresponds to empirical coverage equal to α for every α. Figure 7 shows the result.

Two findings emerge. First, both interpretations overshoot nominal coverage at every level: empirical coverage is systematically higher than α under either distribution. At the widely cited $\pm 1\sigma$ point ($\alpha = 0.683$), empirical coverage under the Gaussian interpretation is 0.994; at $\alpha = 0.5$, empirical coverage under the Laplace interpretation is 0.824. Second, and more informatively, the Laplace interpretation is systematically *less* miscalibrated than the Gaussian interpretation at every α.

This ordering is consistent with the form of the training loss. DESS is trained with an L1 loss on the spread parameter, targeting $\beta \cdot |\delta|$ where δ is the mean-estimation error. Under L1, $\hat{\boldsymbol{\sigma}}$ is the maximum-likelihood estimator of the scale parameter b of a Laplace distribution, which relates to a Gaussian standard deviation by

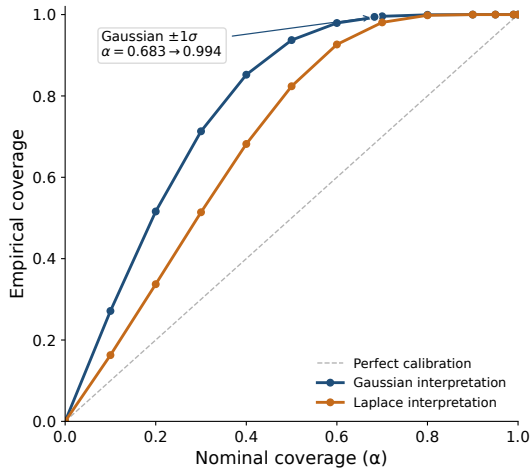

Fig. 7 Calibration of the DESS spread parameter under two interpretations on 20 Newsgroups. Each curve plots empirical coverage (fraction of test points falling within the nominal interval) against nominal coverage level α. The dashed grey diagonal is perfect calibration. Interpreting $\hat{\boldsymbol{\sigma}}$ as a Gaussian standard deviation (blue) overshoots substantially across all α, including the widely cited $\pm 1\sigma$ point (annotated), where empirical coverage is 0.994 rather than the nominal 0.683. Interpreting $\hat{\boldsymbol{\sigma}}$ as a Laplace scale parameter (orange) is less miscalibrated but still overshoots. The ordering Gaussian > Laplace (in miscalibration) is consistent with the L1 training loss, under which $\hat{\boldsymbol{\sigma}}$ is the MLE of a Laplace scale, and has direct implications for the choice of inference-time sampling distribution analyzed in Section 5.7.

$\sigma_{\text{Gauss}} = b\sqrt{2}$ [23]. A back-of-the-envelope calculation combines this conversion with the $\beta = 1.5$ multiplier at training time: if $\hat{\boldsymbol{\sigma}}$ were a calibrated Laplace scale, its Gaussian interpretation would correspond to an inflation factor of $\beta \cdot \sqrt{2} \approx 1.5 \cdot 1.414 \approx 2.12\times$, placing empirical coverage at $\alpha = 0.683$ near 0.96. The observed value of 0.994 exceeds this prediction, suggesting that additional scale is introduced at training time beyond what the simple back-of-envelope captures. A likely contributor is the multi-target dispersion term in Equation 6: with $\alpha = 0.5$ and $T = 16$ targets per class, the per-dimension range Δy is systematically larger than the residual δ, inflating $\hat{\boldsymbol{\sigma}}$ beyond the single-target Laplace-MLE prediction. We leave a precise decomposition of this inflation to future work.

The practical consequence matters more than the exact theoretical decomposition. The *ordering* of miscalibration Gaussian worse than Laplace is robust across the full coverage range, and this ordering has direct implications for how $\hat{\boldsymbol{\sigma}}$ should be used at inference time. If $\hat{\boldsymbol{\sigma}}$ were interpreted as a Gaussian σ for sampling, the sampling scale would be inflated by more than if it were interpreted as a Laplace scale b. The choice of inference-time sampling distribution is therefore not arbitrary; we turn to it next.

5.7 Alternative Sampling Distributions

We compare Gaussian, Laplace, and Uniform sampling while keeping the learned spread fixed — one trained DESS model, three inference-time interpretations. The choice of distribution has substantial downstream consequences. Under the evaluated sampled-classification setting (at $\beta = 1.5$), Uniform sampling performs best (accuracy = 0.3592), followed by Gaussian (0.1565), while Laplace performs worst (0.0722). All three modes degrade substantially relative to the point-estimate accuracy of 0.70; the comparison is between different degrees of degradation, not between preservation and collapse.

The calibration analysis of Section 5.6 provides a principled interpretation of this ordering. The DESS spread head behaves more like a Laplace scale parameter than a Gaussian standard deviation, but both interpretations overshoot the nominal sampling scale, with Gaussian overshooting more than Laplace. Laplace sampling (heavier tails than Gaussian) further amplifies this inflation at inference time. Uniform sampling, in contrast, has bounded support: samples are strictly confined to $[\hat{\boldsymbol{\mu}} - \hat{\boldsymbol{\sigma}}, \hat{\boldsymbol{\mu}} + \hat{\boldsymbol{\sigma}}]$ along each dimension, which truncates the inflated tails and keeps samples closer to the class manifold.

In summary, the ordering Uniform > Gaussian > Laplace reflects *increasing mismatch* between inference-time distribution and the Laplace-like shape of the training-time posterior. This result should not be overgeneralized into a universal claim about distribution choice. It does, however, support the interpretation that DESS is best thought of as a family of sampling mechanisms operating on a learned uncertainty structure, not merely as a fixed Gaussian recipe, and that the distribution used at inference time should be chosen to match the training-time posterior rather than assumed to be Gaussian by default.

5.8 Summary

Experiment 2 establishes a more nuanced picture of DESS than the synthetic experiment of Section 4 alone could support. The predicted spread parameter is meaningful in the sense that it tracks empirical within-class variance with high rank correlation, and the spread hyperparameter β provides a controllable relevance–diversity tradeoff under sampling. The behavior of β in high-dimensional embedding spaces is governed by concentration of measure, which we turn into a predictive design rule that transfers across datasets. The learned spread is calibration-shaped as a Laplace scale parameter rather than a Gaussian standard deviation, with

consequences for the choice of inference-time sampling distribution. At the same time, the spread norm is not by itself a useful out-of-distribution signal, and DESS does not outperform a strong softmax classifier on direct accuracy. We turn next to a realistic downstream task to test whether DESS's behavior in this controlled analysis carries over to a benchmark with established baselines.

6 Experiment 3: DESS for Candidate-List Reranking on SQID

Experiments 1 and 2 examined DESS in increasingly realistic settings where ground truth is either known by construction (Section 4) or recoverable from class structure (Section 5). Both established that DESS's spread parameter is geometrically meaningful and that its behavior in high-dimensional embedding spaces is governed by concentration of measure. The natural next question is whether these findings translate into measurable gains on a benchmark-driven downstream task with strong established baselines. This experiment addresses that question.

6.1 Benchmark and Protocol

We use the Shopping Queries Image Dataset (SQID) [15] together with the ESCI Task 1 reranking setup [16]. We follow the benchmark protocol faithfully:

1. only the judged candidate list attached to each query is reranked,
2. no full-catalog retrieval is performed,
3. evaluation uses corrected gains $E = 1.0$, $S = 0.1$, $C = 0.01$, and $I = 0.0$ (following the correction documented in [15]), and
4. the headline metric is NDCG.

This protocol matters for two reasons. First, it provides a clear real-world benchmark where ranking quality is measured directly against human-judged relevance. Second, it prevents the evaluation from drifting into a different task than the one used by the SQID paper: the goal is to test whether DESS helps under the actual reranking protocol, not under a redesigned benchmark. The dataset slices used in our experiments contain 419,653 judged training rows across 20,888 queries and 181,701 test judgements across 8,956 queries.

6.2 Reproduction Baselines

Before adding DESS, we reproduce the key SQID reranking baselines. Table 4 shows the main results.

Table 4 SQID reranking reproduction baselines under corrected NDCG.

Method	NDCG
Random	0.7465
CLIP-text [27]	0.8106
CLIP-image [27]	0.8223
SBERT-text [12]	0.8294
Best fusion (SBERT + CLIP-image)	0.8402
ESCI baseline cross-encoder [16]	0.8550

These numbers establish the reference frame for the DESS evaluation. A competitive DESS reranker should at minimum remain in the range of the stronger text-based baselines. Surpassing the cross-encoder is a much harder target.

6.3 DESS Reranker Variants

For this experiment, we keep the benchmark fixed and add DESS as a separate reranker on top of frozen sentence-embedding features produced by `all-MiniLM-L12-v2` [12, 13]. This is a deeper variant of the `all-MiniLM-L6-v2` encoder used in Experiment 2; using two different backbones across the two experiments provides an implicit robustness check that DESS's behavior is not tied to a specific encoder checkpoint. We evaluate four DESS variants:

1. *mlp joint*: one three-layer MLP jointly predicts mean and spread,
2. *frozen mu sigma mlp*: the frozen query embedding is kept as $\hat{\boldsymbol{\mu}}$ and only $\hat{\boldsymbol{\sigma}}$ is learned,
3. *dual head detached sigma*: separate mean and spread heads, with the spread head conditioned on a detached mean, and
4. *dual head query concat sigma*: the spread head sees both the query embedding and the detached mean.

These variants test a practically important design question: should DESS reshape the semantic center of the query embedding, or should it keep a strong frozen embedding as the central point and learn only the surrounding uncertainty? The findings of Experiment 2 suggest a hypothesis: variants that move the mean in a high-dimensional space co-learn both center and spread, which interacts unfavorably with concentration of measure (Section 5.4). The conservative variant, which keeps the pretrained mean frozen and learns only $\hat{\boldsymbol{\sigma}}$, should perform best.

6.4 Reranking Results

Table 5 and Figure 8 show the main DESS reranking results.

Table 5 DESS reranker results on SQID.

Variant	NDCG	Δ vs. SBERT	Δ vs. fusion
frozen mu sigma mlp	0.8296	+0.0002	−0.0106
dual head detached sigma	0.8078	−0.0216	−0.0325
dual head query concat sigma	0.8076	−0.0218	−0.0326
mlp joint	0.8035	−0.0259	−0.0367

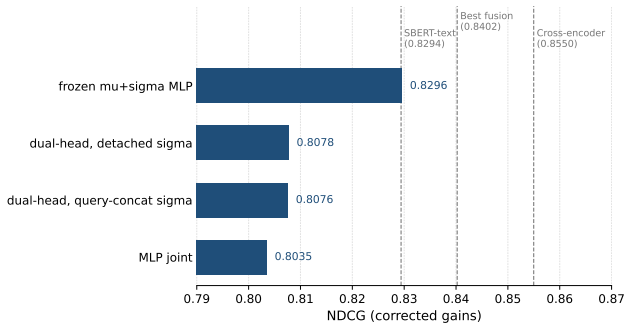

Fig. 8 NDCG on the SQID/ESCI Task 1 benchmark for the four DESS reranker variants introduced in Section 6.3 (bars), with external baselines from Section 6.2 shown as reference lines (grey dashed): SBERT-text (0.8294), best multimodal fusion with SBERT + CLIP-image (0.8402), and the ESCI cross-encoder baseline (0.8550). The best DESS variant (*frozen mu sigma mlp*) retains the frozen query embedding as the mean and learns only the per-dimension spread around it, matching the SBERT-text baseline within 0.0002 NDCG. Variants that learn the mean via an MLP fall behind. All DESS variants remain below the best multimodal fusion and cross-encoder baselines.

The pattern is clear and matches the hypothesis from Section 6.3. The best DESS variant is the one that keeps the frozen SBERT query embedding as $\hat{\boldsymbol{\mu}}$ and learns only $\hat{\boldsymbol{\sigma}}$. This variant reaches 0.8296 NDCG, essentially matching the 0.8294 of the SBERT-text baseline. All variants that move the mean with an MLP perform worse.

This is a meaningful negative result. It suggests that, at least for this reranking task, DESS is best used to model uncertainty around a strong pretrained representation rather than to replace that representation with a new learned center. Put differently, DESS is more reliable as an uncertainty-aware extension of a good embedding model than as a new semantic encoder in its own right.

The same pattern appears in the sampling-based reranker sweep. Over three seeds, the sampling version of *frozen mu sigma mlp* reaches a mean NDCG of 0.8273 ± 0.0003, again clearly ahead of the other DESS variants but still below the best fusion and cross-encoder baselines.

This outcome connects back to the design rule of Section 5.5, though the connection is one of spirit rather than direct applicability: the design rule was derived under class-conditional single-target training, and SQID reranking is a query-conditional pairwise task, so the rule's exact prediction does not transfer to this setting. Nonetheless, *frozen mu sigma mlp* leaves the strong SBERT mean intact and learns only a per-dimension uncertainty around it — which is close in spirit to the kind of use-case the rule is designed for: small, geometry-respecting spread around a trustworthy centroid. Variants that also move the mean are in effect co-learning both the center and the spread of a distribution in a high-dimensional space where, as Experiment 2 showed (Section 5.4), even modest spread displaces samples by $\sqrt{d} \cdot \bar{\sigma}$. Their lower NDCG is consistent with that geometric hazard.

6.5 Query-Level Behavior

Aggregate NDCG tells only part of the story. Query-level inspection shows that the best DESS variant helps some queries substantially while hurting others.

Among the largest improvements relative to the SBERT-text baseline are queries such as "plastic cup drink sleeve with handle", "kaizen home goods", and "butterfly house for outside". Among the largest regressions are more specific or SKU-like queries such as "iphone xr usb" and "contact lens".

This pattern is consistent with the broader interpretation of DESS across this paper. The method appears most useful when a query admits a meaningful neighborhood of plausible products in embedding space, so that uncertainty around the central representation can help reweight candidates. It appears less useful when the query is highly specific and the task rewards a sharp, nearly deterministic semantic match. This mirrors the synthetic-experiment finding (Section 4.2) that smaller β favors relevance and larger β favors diversity — DESS is doing relevance–diversity reweighting, and diversity helps only when the candidate set genuinely contains relevant alternatives.

6.6 Summary

Experiment 3 places DESS in a benchmark-driven setting and produces a useful negative result. The best DESS variant is competitive with a strong frozen text-only baseline but does not surpass cross-encoder or multimodal fusion baselines. The variant that wins is the one that respects the geometric intuition developed in Experiment 2: keep the pretrained mean as the central point and learn only a per-dimension spread around it. Variants that try to relearn the embedding center underperform. Query-level analysis suggests that DESS

helps queries where uncertainty over a neighborhood of plausible products is meaningful, and hurts queries where deterministic semantic matching is required.

Together with Experiments 1 and 2, this completes the empirical picture of DESS that the paper presents: a parameter-efficient probabilistic sampling framework whose spread parameter is geometrically meaningful and whose downstream usefulness depends on the structure of the task. The discussion section that follows takes stock of what these findings say about the method as a whole.

7 Discussion

The three experiments together support a more precise picture of DESS than any one of them in isolation. We collect the main lessons here.

DESS recovers structured uncertainty across settings of increasing realism. In Experiment 1 (Section 4), DESS recovered ground-truth distributional parameters on controlled Gaussian data and exposed β as a controllable lever for the relevance–diversity tradeoff. In Experiment 2 (Section 5), the predicted spread continued to carry meaningful signal: predicted $\hat{\boldsymbol{\sigma}}$ and empirical within-class variance correlated at Spearman $\rho = 0.94$ across all (class, dimension) pairs, even though the embedding space was learned from noisy text data and the model received no direct supervision on the per-dimension scale. In Experiment 3 (Section 6), the variant that respected the geometric intuition of the previous experiments — keep the pretrained mean, learn only the spread — was also the variant that matched the strong SBERT-text baseline. Across all three settings, the spread parameter is not arbitrary: it tracks something real about the data.

Low-dimensional intuitions about β do not transfer to high dimensions. The $\beta = 1.5$ value identified in Experiment 1 performed poorly in Experiment 2 because of concentration of measure [17]: in $d = 384$, the same β produces an L2 displacement that exceeds the inter-centroid distance in the embedding space, and nearest-class retrieval breaks. We turned this observation into a geometric design rule (Equation 8) that predicts a safe operating range for β from training-set geometry alone, and validated the rule on a second dataset to within 20% of the empirical threshold. We see this as one of the more practically useful contributions of the paper: practitioners deploying DESS on a new embedding space can compute a starting β from the data without a training sweep, and the rule generalizes the principle that sample displacement should remain below the resolution of the target manifold.

DESS is best understood as an uncertainty layer, not a direct accuracy booster. On 20 Newsgroups classification (Experiment 2) DESS did not outperform a softmax classifier, and on SQID reranking (Experiment 3) the strongest DESS variant matched but did not surpass the SBERT-text baseline. We interpret this honestly: DESS is not a universal replacement for point-estimate models. What it adds is an explicit uncertainty structure in the embedding space — which, in applications that genuinely benefit from probabilistic sampling (diverse retrieval, probabilistic recommendation, content selection with controllable variability), is not something a point estimate can provide. DESS's value should be measured against tasks that need that structure, not against tasks that reward sharp deterministic matching.

The form of uncertainty matters. The calibration analysis of Section 5.6 surfaced a result we did not initially expect. The DESS spread parameter is not well-described as a Gaussian standard deviation: empirical coverage at the nominal $\pm 1\sigma$ point is 0.994 rather than 0.683. It is better described as a Laplace scale parameter, in line with the L1 loss used to train $\hat{\boldsymbol{\sigma}}$. Even under the Laplace interpretation, $\hat{\boldsymbol{\sigma}}$ overshoots nominal coverage at every level, suggesting that the multi-target dispersion term in the loss (Equation 6) introduces additional scale beyond the single-target Laplace-MLE prediction. The practical consequence is concrete: the inference-time sampling distribution should be chosen to match the training-time posterior shape rather than defaulted to Gaussian. Bounded sampling (Uniform) outperformed both Gaussian and Laplace in our setting, because it truncates the inflated tails most aggressively.

Per-dimension uncertainty is not the same as out-of-distribution detection. Despite the strong correlation between $\hat{\boldsymbol{\sigma}}$ and within-class variance, the norm $\|\hat{\boldsymbol{\sigma}}\|$ was indistinguishable from chance as an out-of-distribution signal (AUROC 0.49). This is consistent with the aleatoric/epistemic distinction [20, 22]: DESS's spread is shaped by within-cluster variation that the model has seen, not by epistemic uncertainty about clusters it has not. Practitioners hoping to use DESS for OOD detection or selective abstention will need to pair it with an explicit uncertainty objective or an auxiliary calibration mechanism. The spread parameter alone is the wrong tool for that job.

Preserve the pretrained mean when possible. Experiment 3 provided a clean architectural lesson: among the four DESS variants tested for SQID reranking, the one that retained the frozen SBERT embedding as $\hat{\boldsymbol{\mu}}$ and learned only $\hat{\boldsymbol{\sigma}}$ won. Variants that learned a new mean via an MLP underperformed. We read this as a more general design heuristic: when a strong pretrained representation is available, DESS is more useful as a small, geometry-respecting uncertainty layer on top of that representation than as a new semantic encoder competing with it. This heuristic is consistent with the design rule of Section 5.5, which favors small spreads around trustworthy centroids in high-dimensional spaces.

Some retrieval failures are downstream-policy failures, not DESS failures. We observed in exploratory experiments that naive sampling from DESS can collapse onto repeated nearest neighbors unless the downstream selection procedure is made explicitly diversity-aware. This is a property of nearest-neighbor lookup combined with concentrated samples, not a defect of the spread parameter. Downstream decoding and selection rules — from temperature scaling to maximum marginal relevance [3] to listwise reranking — matter as much as the learned uncertainty structure. DESS provides the structured uncertainty; how that uncertainty is consumed is a separate engineering question.

8 Limitations and Future Work

We identify several limitations of the present work, organized around the components of DESS that are most likely to need revisiting in follow-up research.

Per-dimension independence. DESS factors $p(y \mid x)$ as a product of per-dimension univariate densities (Equation 2), which trades expressiveness for trainability. In the embedding spaces we studied, this approximation worked well empirically: the predicted spread tracked empirical variance at $\rho = 0.94$. But embedding dimensions in real models are not independent, and the diagonal formulation cannot capture covariance structure. A direction worth exploring is a low-rank correction to the diagonal formulation that captures dominant covariance modes without requiring the full $O(d^2)$ parameterization of mixture density networks.

Uncertainty calibration. Section 5.6 established that DESS's spread parameter is calibration-shaped as a Laplace scale rather than a Gaussian standard deviation, but neither interpretation is perfectly calibrated — both overshoot nominal coverage. The back-of-envelope calculation explained part of the overshoot, with the remainder plausibly attributable to the multi-target dispersion term. A precise decomposition of the inflation factor and a calibration-aware training objective (e.g., a calibration penalty added to $\mathcal{L}_\sigma$) are natural next steps. This work would also bear on the OOD-detection limitation: an explicit uncertainty objective could potentially recover epistemic information that the current spread parameter does not capture.

Scope of the geometric design rule. The design rule of Equation 8 was derived under single-target class-conditional training and validated on two class-conditional datasets. We empirically verified that the rule remains conservative under the multi-target loss with $\alpha = 0.5$, but its exact form in other training regimes — query-conditional pairwise tasks, listwise ranking, or other settings where the target is not a class but a candidate list — is not established. The general principle (sample displacement must remain below the resolution of the target manifold) should carry over, but the specific functional form may not.

Multimodal and listwise extensions. Experiment 3 used DESS as a text-only reranker on SQID, leaving the dataset's multimodal information unused. The strongest non-DESS baseline in our reproduction table was a text-image fusion model. A multimodal DESS reranker operating over joint text-image representations is the most direct way to test whether DESS's uncertainty structure complements multimodal features rather than substituting for them. Relatedly, the current reranking formulation adapts DESS to pairwise query-product training data; a listwise or candidate-set-aware DESS objective would more directly exploit the structure of the reranking task.

Query-dependent spread control. Throughout this paper we treated β as a global hyperparameter, fixed for the entire training run. A more flexible variant would let the model produce β as a per-input output, allowing the predicted spread to scale with query difficulty or specificity. This is a natural fit for the SQID query-level findings of Section 6.5, where DESS helped broad queries and hurt highly specific ones — a per-query β might let the model collapse the distribution for SKU-like queries while broadening it for exploratory ones.

Connection to retrieval diversification. The discussion of sampling-distribution mismatches (Section 5.7) and the observation that naive nearest-neighbor lookup can collapse onto repeated samples both point at the same

underlying gap: DESS provides a learned uncertainty structure, but how that structure is consumed by downstream retrieval is a separate engineering layer. Tighter integration between DESS uncertainty estimates and explicit diversification mechanisms (maximum marginal relevance and its successors) is a promising direction.

9 Conclusion

This paper presents DESS (Dimensional-PDFs for Embedding Space Sampling), a parameter-efficient framework for probabilistic sampling in learned embedding spaces. DESS attaches to any backbone with two output heads, trains with simple element-wise losses and no sampling in the loop, and produces a per-dimension probability density that can be sampled at inference to retrieve from an output set. We validated the method through three experiments of increasing realism: controlled synthetic Gaussian data, sentence-transformer embeddings of 20 Newsgroups, and candidate-list reranking on the SQID/ESCI Task 1 benchmark.

The central findings are these. The DESS spread parameter is geometrically meaningful: its predicted values track empirical within-class variance with Spearman $\rho = 0.94$ on real text embeddings, even though the model receives no direct supervision on spread magnitude. The behavior of the spread hyperparameter β in high-dimensional embedding spaces is governed by the concentration of measure, and we derived a geometric design rule (Equation 8) that predicts a safe operating range for β from training-set geometry alone, validated across two datasets. The learned spread is calibration-shaped as a Laplace scale parameter rather than a Gaussian standard deviation, with consequences for the choice of inference-time sampling distribution: bounded sampling preserves accuracy under sampling more effectively than infinite-support distributions. On the SQID benchmark, the DESS variant that respects the design rule's spirit keep the pretrained mean, learn only the spread matches a strong SBERT-text baseline.

DESS is not a universal replacement for point-estimate models. It does not outperform softmax classification on direct accuracy, and on SQID reranking it does not surpass cross-encoder or multimodal fusion baselines. Its spread parameter is also not by itself a useful out-of-distribution signal. We argue that the most defensible interpretation of DESS is not as a competitor to discriminative methods but as a lightweight uncertainty-aware layer for embedding-space models one whose spread parameter is geometrically constrained, probabilistically Laplace-shaped, and most effective when used to model variation around a strong pretrained mean. That role is already useful, and the experiments in this paper characterize the conditions under which it holds and the conditions under which it does not.

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